

Question:

At birth, the least developed part of the brain is the

Answer Choices:

- A. visual system
- B. thalamus
- C. frontal lobe
- D. brain stem
- E. parietal lobe
- F. cerebellum
- G. cortex
- H. basal ganglia
- I. hypothalamus
- J. limbic system

Reasoning + Answer:

<think>

The brain’s overall developmental trajectory shows consistent , predictable patterns : neuronal migration and synaptic maturation progress from caudal to rostral regions , and myelination (an important determinant of speed and efficiency) similarly advances cranially . Specifically , the telencephalon (cortex and associated structures) matures more slowly than structures that arise earlier in ontogeny , such as the brain stem and cerebellum .

By birth , the brain stem (medulla , pons , mid brain) and cerebellum (medially , fast-spinning climbing fibers innervating deep cerebellar nuclei ; broadly , cortical foliation and connections) show substantial myelination and synaptic refinement . The thalamus has largely finished migration and de novo synapses , and parietal cortex is relatively mature because it emerges early in cerebral development . By contrast , frontal cortex shows prominent ‘glutamic acid decarboxylase’ staining at term , reflecting its primary role in controlling myelination , and posterior cingulate / precuneus often forms only later in postnatal life . Despite macroscopic appearance at birth , the cortex itself remains the last major region to undergo extensive synaptic elaboration , myelination , and structural refinement ; clinically , neonates and infants express heightened plasticity and maturational lag in cognitive and motor functions .

Therefore , the least developed part of the brain at birth is the cortex .

Correct answer : G . cortex

</think>

<answer>

G . cortex

</answer>

